

BOWEN YANG

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EDUCATION

Shanghai Jiao Tong University (SJTU)

B. Eng in Information Engineering, B.A in French Studies

Graduating Expected June 2027

SJTU Paris Elite Institute of Technology

GPA: **91.7/100**; Rank: **7/110**

Advisor: **Prof. Junchi Yan, Prof. Xiaosong Jia**

RESEARCH EXPERIENCE

GuidedVLA: Specifying Task-Relevant Factors via Plug-and-Play Action Attention Specialization *Robotics: Science and Systems (RSS) 2026*

Xiaosong Jia*, **Bowen Yang***, Zuhao Ge*, Xian Nie*, Yuchen Zhou*, Cunxin Fan*, *et al.*, Hongyang Li, Zuxuan Wu, Junchi Yan, Yu-Gang Jiang

**Equal contribution; paper to appear*

Research Assistant, RethinkLab, SJTU

Sep 2025 – Jan 2026

- Led the **codebase implementation and simulation experiments** for GuidedVLA.
- Implemented **specialized action-attention heads** for object grounding, temporal skill recognition, and depth-based geometric perception in the π_0 action decoder.
- Integrated a **ControlNet-style residual adapter** for plug-and-play factor supervision while preserving the pretrained VLA policy pathway, enabling stable guidance injection without disrupting the base policy.
- Boosted average success rates from **68.2% to 75.4%** on **LIBERO-Plus** and **77.4% to 90.6%** on **RoboTwin 2.0**, while improving training throughput by **30%+** via mixed precision and **torch.compile**.

ROBOTICS COMPETITIONS & PROJECTS

RoboMaster University Championship

Vice-Captain & Algorithm Lead, Team Jiao Loong

Oct 2023 – Aug 2025

- As **Vice-Captain and Algorithm Lead**, managed the R&D of all algorithm projects across SLAM, navigation, perception for Team Jiao Loong, and won **National Champions in 2024 & 2025**.
- Led development of a **LiDAR-inertial SLAM pipeline** (Fast-LIO2 based) with GICP map registration and fully PTP-synchronized LiDAR timing to improve robustness under fast robot motion.
- Built a **motion planning system** combining **kinodynamic A*** and **MPC** and achieve low-latency tracking on embedded platforms under real-world competition constraints.
- Deployed **TensorRT-accelerated YOLO** on **Jetson Orin** and integrated an **EKF-based motion prediction** module for reliable enemy-state estimation under occlusion and high-speed dynamics.

AWARDS AND HONORS

- **National Scholarship** 2024, 2025
Ministry of Education of China; top 0.2% of students nationwide.
- **National Champion**, RoboMaster University Championship 2024, 2025
Core contributor to SLAM, navigation, perception, and embedded systems.
- **RM Award (Algorithm)**, RoboMaster University Championship 2025
Individual award for SLAM and navigation algorithm design of the Sentry robot.
- **Finalist (Top 2%)**, Mathematical Contest in Modeling (MCM) 2025
Problem B (Discrete). Environmental Dynamic System Evolution and Multi-Objective Optimization Modeling
- **NOI Winter Camp Bronze; USACO Platinum Division** 2020–2021
Advanced training in algorithms and competitive programming.

TECHNICAL SKILLS

Programming: Python, C, C++

Frameworks & Tools: PyTorch, TensorFlow, ROS/ROS2, OpenCV, TensorRT, Docker, Git, $\LaTeX$

Languages: English (TOEFL R6/L5.5/S6/W5), French (DELFB2), Cantonese (Conversational)